SHAHRAM AJLOO

Electrical & Instrumentation · Instrumented design, commissioning and validation

+45 93 99 95 14 · ajlooshahram@gmail.com · linkedin.com/in/shahramajloo · Måløv, Denmark

5 November 2026

NNE A/S · Kalundborg · Electrical & Instrumentation Engineer (req5138)

Application: Electrical & Instrumentation Engineer

Dear Søren Hedegaard

An instrument is a promise. That the temperature on the screen is the temperature in the pipe, that the valve moves when the loop tells it to, and that the safety function works when everything else has failed. In an API plant that promise is what separates a released batch from a rejected one — and keeping it is the whole job of an E&I engineer: specify it correctly, document it honestly, test it in the field and prove it in the qualification. That is the work I want to do at NNE.

I work from four fixed points. First, design out from the P&ID: instrument index and datasheets, measuring principle chosen for the medium and the range, materials and process connection, hazardous area and zone, and — where the function is safety-related — the SIL target and the SIS architecture that follows from it. Second, verify in the field: loop checks against the IO list, calibration, stroke and response tests on actuators, and functional testing of the interlock before anyone signs anything. Third, validation with evidence: IQ/OQ/PQ with a traceability matrix from requirement to test, deviations closed with a cause and a verified corrective action, and change control before the change rather than after it. Fourth, the vendor and construction interface: agreed scope, agreed interface list, FAT before shipment, and supervision on site where the work happens — because that is where the detail that nobody specified becomes a problem.

I have done this inside Novo Nordisk, your customer. From 2022 to 2024 I was senior commissioning engineer and project manager for electrical systems in Kalundborg and Bagsværd: 37 parallel capital projects, 25 specialists and DKK 2–10m per project, with acceptance criteria, IO and instrument documentation, panel and cable schedules, loop checks, FAT/SAT, IQ/OQ/PQ, deviation handling and audited handover — zero critical findings under GMP, FDA and EMA. I have stood in the design review, in the panel workshop and on the steel with the installer, and I have defended the documentation in front of an inspector.

The hands-on side is not a gap in my CV, it is where I come from. Five years as maintenance and operations manager at Biofos on a 24/7 plant: sensors, transmitters, actuators, pumps, HVAC and high-voltage switchgear, alarms and drift handled from SCADA data, calibration and fault diagnosis on the spot, and LOTO and permits before anything was touched. Since 2023 I have run ELOHVAC with five electricians and three apprentices — panels, motor controls, variable speed drives and control systems for heating, ventilation and cooling — where the specification, the installation, the test and the customer are the same day's work. Before that, twelve technicians and a machine workshop at ARJ.

The technical foundation fits what you ask for. I am an **authorised marine engineer** — the degree you list — with Danish electrical authorisation at LV, MV and HV, and a B.Tech in Technology Management & Mechanical Engineering with electrical engineering studies behind it. Marine engineering is exactly the discipline that teaches you to trust a measurement: pressure, temperature, flow, level and vibration, the instrumentation that reads them, and the control loop that acts on them. I read the P&ID, the single-line diagram, the instrument datasheet and the loop drawing, and I know which of the four usually disagrees with the other three. A department of a hundred engineers is also a place where standards travel: what one project documents well, the next one reuses — and I would rather work where the discipline is bigger than one person.

NNE's values are the ones I already work by. **Take initiative and get things done**: an open point in the field is closed the same week, not carried to the next milestone. **Be passionate about pharma engineering expertise**: the datasheet and the test record are the product. **Innovate and create efficient solutions**: a reusable checklist or template is worth more than a heroic evening. **Team up and leverage our global knowledge**: the answer often already exists in another discipline or another country. And **exceed customer expectations**: the customer is Novo Nordisk, and the patient is the reason the cable is pulled correctly. I also value the way you describe the start here — an onboarding that is taken seriously, a flat hierarchy and colleagues who answer questions.

Practically: I live in Måløv, about an hour from Kalundborg, and I will be on site five days a week — in this discipline the field is where the work is. Danish is my native language and English my professional one, so I can work in both without switching down. I can start by agreement, and I would welcome a conversation about which projects the E&I department is staffing over the next year, and where a commissioning and validation background is most useful to you first. If it is useful, I can also show how I document a loop from requirement to signed test — that is the part of the job I can demonstrate, not just describe.

Kind regards

Shahram Ajloo

SHAHRAM AJLOO

Electrical & Instrumentation · Instrumented design, commissioning and validation

+45 93 99 95 14 · ajlooshahram@gmail.com · linkedin.com/in/shahramajloo · Måløv, Denmark

PROFILE

E&I engineer with 20+ years from electrical and instrumentation work, regulated pharma production and 24/7 technical operations — authorised marine engineer with Danish electrical authorisation (LV/MV/HV) and a B.Tech in Technology Management & Mechanical Engineering. Two years at Novo Nordisk with 37 parallel capital projects, loop checks, FAT/SAT and IQ/OQ/PQ and zero critical findings at handover; five years of hands-on operations with sensors, transmitters, actuators and SCADA at Biofos; three years building panels, drives and control systems at ELOHVAC.

INSTRUMENTATION, COMMISSIONING AND VALIDATION

- **Instrumented design:** Instrument index and datasheets, measuring principle and range, materials and process connections, hazardous-area considerations and safety functions — designed out from the P&ID and checked against the process before anything is bought.
- **Field verification and commissioning:** IO and loop checks against the list, calibration and stroke tests on actuators, functional testing of interlocks and sequences, SAT and handover — documented while the work happens, not reconstructed later.
- **Validation and qualification:** URS and design review, risk assessment and FMEA, IQ/OQ/PQ, traceability matrix, deviation and change control; audited handover with zero critical findings across 37 GMP projects at Novo Nordisk.
- **Control systems and data:** PLC- and SCADA-based systems and BMS/FMS: alarm limits and escalation, trends and condition data, interface protocols such as Modbus and BACnet; fault finding from sensor to screen.
- **Site supervision and coordination:** Construction coordination with contractors and vendors on live plants: scope and interface lists, FAT before shipment, permits, LOTO and safe work practices, and daily coordination when the work is running.

TOOLS AND METHODS

Task	Tool and documented use
Electrical and instrumentation	P&IDs, single-line diagrams, instrument datasheets and loop drawings, panel and cable schedules, IO lists, ATEX and hazardous-area requirements; electrical authorisation LV/MV/HV.
Functional safety and standards	Safety functions and SIS architecture, SIL targets, interlock testing and documentation to industrial standards and authority requirements.
Testing and qualification	FAT/SAT, IQ/OQ/PQ, calibration and loop checks, acceptance criteria and traceability matrix; deviations closed with root cause and verified corrective action.
Operations and maintenance	Preventive maintenance from condition data, spare-part criticality, calibration routines, LOTO, workplace assessment and permits in a 24/7 operation.
Vendors and construction	Vendor scope and interface lists, FAT and inspection before shipment, supervision and construction coordination on site with contractors and client representatives.
Digital and data	Python, Claude API, SAP and Excel: traceability matrices, IO and deviation logs, alarm statistics and dashboards — over 40 % of administrative work removed from quoting and follow-up.

EXPERIENCE

Owner / Technical Director — ELOHVAC ApS · Copenhagen, Denmark · May 2023–present

Scope: 5 electricians and 3 apprentices · full P&L · commercial and industrial customers

- **Panels, drives and controls:** Electrical panels, motor controls, variable speed drives and control systems for heating, ventilation and cooling — specified, installed, tested and handed over with documentation.
- **Customers and people:** Recruitment, 1:1s and apprentice training alongside budget, quoting and margin; customers informed the same day something changes.
- **Digital operations:** Python and Claude API for quotations, scheduling, checklists and KPI dashboards; over 40 % productivity gain in the business.

Senior Commissioning Engineer / Project Manager, electrical systems — Novo Nordisk A/S · Kalundborg / Bagsværd, Denmark · Nov 2022–Jul 2024

Scope: 37 parallel capital projects · 25 specialists · DKK 2–10m each · GMP/FDA/EMA

- **Commissioning and qualification:** Acceptance criteria, FAT/SAT, loop checks and IQ/OQ/PQ with traceable documentation and deviation handling; zero critical findings at handover.
- **Documentation and reporting:** IO and instrument documentation, panel and cable schedules and as-built records; progress, cost, risk and resources reported monthly to management and client.

Project and Workshop Manager — Ventek-Air · Copenhagen, Denmark · Jan 2022–Nov 2022

Scope: 14 employees · 11 parallel construction projects · DKK 2–10m each

- **Installation and handover:** Tenders, calculations, subcontractors and handover packages in ventilation and technical installations; daily board meetings and bottleneck analysis.

Maintenance and Operations Manager / Workshop Technician — Biofos, Lynetten · Copenhagen, Denmark · Aug 2017–Jan 2022

Scope: 24/7 plant · sensors, actuators, pumps, HVAC, high-voltage switchgear, SCADA

- **Hands-on operations and fault finding:** Sensors, transmitters, actuators and switchgear in daily operation: calibration, troubleshooting, corrective work and alarm handling from SCADA data in shift teams.
- **Preventive maintenance and safety:** Plans from condition data, spare-part criticality and LOTO, workplace assessment and permits before every job.

Electromechanical Technician / Production Team Leader — ARJ Engineering · Tehran, Iran · Dec 2003–Mar 2010

Scope: 12 mechanics and technicians · machine workshop · industrial equipment

- **Equipment and team:** Shift planning and target follow-up; fault diagnosis, root-cause analysis and corrective actions on production equipment.

EDUCATION AND AUTHORISATION

B.Tech, Technology Management & Mechanical Engineering · Copenhagen School of Marine Engineering & Technology Management · 2017–2022 · EQF 6 · GPA 7.6/12

Authorised marine and electrical engineer (LV/MV/HV) · Danish Safety Technology Authority · May 2023

Electrical and mechanical design, electrical engineering · Tehran Technical University · 2000–2003 · EQF 6

Refrigeration and air conditioning · STCW safety and engine-room watch · Mercantec 2021 · Danish Maritime Authority 2021–2023

LANGUAGES, PRACTICAL AND REFERENCES

Danish: native · English: professional, written and spoken · Persian: native · Driving licence B · Lives in Måløv, about an hour from Kalundborg, and on site five days a week · Available by agreement · SAP and Excel in daily use · References from Novo Nordisk and Biofos, and from customers and vendors in my own company, on request.

WHAT THE ROLE ASKS FOR — MY EVIDENCE

Requirement	Evidence
Degree in Electrical Engineering, Marine Engineering, Automation or similar	Authorised marine engineer (maskinmester) with a B.Tech in Technology Management & Mechanical Engineering (EQF 6), electrical engineering studies at Tehran Technical University and Danish electrical authorisation LV/MV/HV.
3+ years in electrical, instrumentation or automation engineering	20+ years: instrument, panel and loop documentation plus FAT/SAT and IQ/OQ/PQ at Novo Nordisk (2022–2024); panels, drives and control systems at ELOHVAC (2023–); SCADA, sensors and actuators at Biofos (2017–2022).
2+ years of hands-on onsite work	Five years of hands-on operations on a 24/7 plant — calibration, fault finding and corrective work — plus three years installing and testing systems on customer sites at ELOHVAC.
Understanding of process control systems and industrial instrumentation	PLC/SCADA and BMS/FMS platforms, control loops, alarm limits and trends; measuring principles for pressure, temperature, flow, level and vibration, and actuator and valve behaviour.
Industrial standards, sensors, instrumentation and P&IDs	Work to industrial standards and authority requirements in both pharma and utility environments; P&ID, single-line diagrams, datasheets and loop drawings as daily documents.
Commissioning, testing or validation of technical systems	FAT/SAT, loop checks, functional testing, IQ/OQ/PQ and audited handover across 37 capital projects, zero critical findings; change control and deviations handled under GMP/FDA/EMA.
Vendor and client collaboration on validation activities	Worked with vendors and client representatives through design reviews, FAT, site testing and qualification; FAT before shipment and agreed interface lists as standing practice.
Professional level in English and Danish	Danish native and English professional, written and spoken; years of work in English-speaking, international project organisations, including teams in Bangalore, India.

FIRST 90 DAYS — DESIGN, FIELD, DOCUMENTATION

Period	What happens, and how it is measured
Week 1–2	Read the live project I am placed on: P&IDs, instrument index, IO list, datasheets and the qualification plan. One day on site with the installers and the client representative. Learn NNE's templates and the project model before adding anything of my own.
Day 30	Delivering on my own scope: instruments specified and datasheets approved, loop checks and functional tests documented in the project system, open points listed with owner and date. One reusable checklist handed to the team.
Day 60	Commissioning and qualification blocks executed and documented for my area, deviations raised and closed with verified corrective actions, and vendor FATs attended with the interface list signed off.
Day 90	I run my area of the discipline independently — design, field verification and qualification — and the project manager can plan against my dates. At least one template or work instruction improved and shared, so the next project starts from a better place than mine did.

THE INSTRUMENT IS A PROMISE

A measurement is only useful if it is right, and it is only right if the whole chain is right: the tapping point, the impulse line, the transmitter, the wiring, the IO card, the scaling and the loop. Most commissioning problems are born in the design, and most design problems are born from an assumption nobody wrote down — the medium is hotter than expected, the line is pressurised differently, the installation is in a zone nobody classified. My habit is to write the assumption down and check it before the purchase order goes out.

SAFETY FUNCTIONS ARE NOT NEGOTIABLE

Where a loop is a safety function, the requirement comes first: what has to happen, how fast, how reliable, and what proves it afterwards. Then the architecture — the sensor redundancy, the logic, the final element — and only then the purchase. The same discipline applies to hazardous areas: zone, equipment category, temperature class and cable entries are not paperwork, they are the reason the plant does not become the incident. I test interlocks against the written requirement, not against what the system happens to do.

COMMISSIONING: THE FIELD DECIDES

A loop is not commissioned when it moves; it is commissioned when it is demonstrated to do the right thing under the conditions the process actually produces. That means IO checked against the list, scaling verified with a simulator, the valve stroked and timed, the interlock tested with the real input, and the result written down at the moment it happened. I have walked out to a panel at the end of a shift to re-check a wiring list before a signature, and it saved the deviation later. The same discipline sits in the documentation: every loop has a record that says what was tested, with what, and by whom.

VALIDATION AND THE AUDIT TRAIL

Qualification is where the E&I engineer proves the design: requirement, risk, test, result, deviation, closure. A traceability matrix that a colleague who was not in the room can follow is worth more than a perfect folder nobody reads. In the pharma projects I have worked in, a missing signature stops a batch release — so documentation is produced with the work, reviewed with the client representative and closed out before the milestone, not rescued afterwards. That is also where trust with the client is built: the test is witnessed, the result is agreed on the day, and the record is closed while the memory is fresh.

VENDORS, SITE AND SUPERVISION

Half of the quality of an instrumented installation is decided before the equipment arrives: a vendor scope that says what is and is not included, an interface list that both sides have signed, and a FAT where the thing is actually tested. The other half is decided on site, which is why construction coordination and supervision belong to the engineer's job: right mounting, right orientation, right cable entries, permits and LOTO in place, and someone who answers when the installer asks a question at 14:30. Keeping the stakeholders — vendor, client representative, contractor and the other disciplines — in the same picture is half of that job, and it is where a project stays calm.

SHARING WHAT WORKS

I build my own tools in Python and Claude — traceability matrices that stay in step with the test list, IO and deviation logs filtered by area and owner, alarm statistics pulled into one page — and I hand them on rather than keeping them — specifications that already carry the right fields, checks that catch the same three mistakes, and continuous improvement that shows up in the next project rather than in a slide. At NNE the same instinct applies across disciplines and countries: what one project learned about a transmitter or a vendor should not have to be learned again in the next project or in Bangalore. That is the knowledge sharing your posting asks for, and it is the part of the job I enjoy most.